

## CS 556 Homework - Vector Spaces and Subspaces

Solve the following questions. Type your solutions using Latex or any other program you prefer and submit your typed solutions as a PDF file.

**Question 1** (10 points) Write the complete solution of the following linear system:

$$\begin{aligned}x + 2y - z &= 1 \\3x + 5y + 2z &= 3 \\2x + y + 13z &= 2\end{aligned}$$

**Question 2** (10 points) Find the rank of the following matrix:

$$\begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 3 & 1 & 1 & 3 \end{bmatrix}$$

**Question 3** (10 points) Construct a matrix A whose column space contains vectors

$$\begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} \text{ and } \begin{bmatrix} 4 \\ 0 \\ 1 \end{bmatrix}, \text{ and whose null space contains the vector } \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

**Question 4** (10 points) Compute the following matrix-vector multiplication as:

- a) Linear combination of columns.
- b) Dot product of rows.

$$\begin{bmatrix} 2 & 1 & 3 \\ 7 & 1 & 0 \\ 3 & 5 & 9 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

**Question 5** (10 points) Find the value of k for which the matrix has:

- a) Dependent columns
- b) Independent columns

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 1 \\ 4 & 8 & k \end{bmatrix}$$

**Question 6** (20 points) Find a basis for the four fundamental subspaces of the matrix.

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$$

**Question 7** (10 points) Consider the vectors  $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$  and  $\vec{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

- a) In the x-y plane mark all nine linear combinations  $c\vec{v} + d\vec{w}$ , with  $c = \{-2, 0, 2\}$  and  $d = \{0, 1, 2\}$ .

b) What shape do all linear combinations  $c\vec{v} + d\vec{w}$  fill? A line? The whole plane? Are the vectors  $\vec{v}$  and  $\vec{w}$  independent?

**Question 8** (10 points) Consider the vectors  $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ ,  $\vec{v} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$  and  $\vec{w} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

a) Can you solve the system  $x\vec{u} + y\vec{v} + z\vec{w} = \vec{b}$ , if  $\vec{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ ?

b) What if  $\vec{b} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ ? How many solutions are there?

c) Are the vectors  $\vec{u}$ ,  $\vec{v}$  and  $\vec{w}$  dependent or independent?

d) Use parts (a) - (c) to decide if  $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$  is an invertible matrix or not.

**Question 9** (10 points) Consider the linear system for some constants b and g:

$$x - 2y + 3z = 3$$

$$2x + y + bz = -4$$

$$x + 0y + 1z = g$$

a) What constant b makes the system singular (missing a pivot).

b) For the value of b found in Part (a), for which values of g, the system has infinitely many solutions?

c) Find two distinct solutions of the system for that g.